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PROJECT TITLE: LIBRARY CIRCULATION AND DEMAND DASHBOARD

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Abstract

The Library Circulation and Demand Dashboard is a menu-driven Python application developed to organise and analyse library book-borrowing records. The project demonstrates how Python programming and data-analysis libraries can be used to process structured circulation data and present useful summaries.

The application stores borrowing transactions in a CSV file and provides options to add, display and search records. It calculates borrowing durations, identifies overdue books, analyses borrowing frequency, and generates category-wise and month-wise summaries. It also performs basic checks for duplicate or incomplete records, creates graphical reports using Matplotlib and exports a summary report in JSON format.

The implementation uses Python for application logic, the CSV and JSON modules for file handling, NumPy for numerical calculations, Pandas for tabular analysis and Matplotlib for visualisation. The application is an academic prototype using demonstration records and can be extended with a database, authentication and a graphical interface.

Keywords:

Python, Library Circulation, CSV, JSON, NumPy, Pandas, Matplotlib, Data Analysis, Data Visualisation.

Table of Contents

Section	Title
1	Introduction
2	Problem Statement and Objectives
3	Software and System Requirements
4	System Design and Methodology
5	Implementation of Major Modules
6	Results and Discussion
7	Testing and Validation
8	Limitations and Future Scope
9	Conclusion
10	References
11	Appendix: CSV Format and Execution

1. Introduction

Libraries maintain records of books issued to students and other members. These records help identify frequently borrowed books, popular categories and books that have not been returned by their due dates. When circulation records are stored only as unanalysed tables, identifying such patterns may require additional manual effort.

The Library Circulation and Demand Dashboard is a small-scale Python application designed to make the analysis of borrowing records easier. It uses a menu-driven interface in which the user selects an operation by entering its corresponding menu number. The system reads and writes records using a CSV file and applies data-processing techniques to generate summaries and reports.

The project is primarily educational. It demonstrates object-oriented programming, file handling, date calculations, data validation, numerical computation, tabular analysis, plotting and report export within one application.

2. Problem Statement and Objectives

Problem statement: Library borrowing records contain useful information, but manually analysing them can be time-consuming. A user may need to find overdue loans, compare borrowing activity across categories, identify frequently borrowed titles or summarise activity by month. Duplicate identifiers and missing fields can also affect the reliability of the records. This project addresses these needs through a simple Python application that processes structured circulation data and presents basic analysis.

The principal objectives are:

- Develop a menu-driven Python application for managing library borrowing records.
- Store and retrieve circulation records using a CSV file.
- Add, display and search borrowing transactions.
- Calculate loan durations using date operations and NumPy calculations.
- Identify books that have not been returned by their due dates.
- Analyse borrowing frequency and category-wise demand.
- Summarise borrowing activity month by month.
- Detect duplicate identifiers, incomplete fields and date inconsistencies.
- Generate graphs for category-wise and monthly borrowing trends.
- Export a structured report in JSON format.

3. Software and System Requirements

Component	Purpose
Python 3	Core programming language used to implement the application.
CSV module	Reads and writes borrowing records in comma-separated value format.
JSON module	Exports a structured report in JSON format.
datetime / date	Validates dates and calculates loan duration and overdue status.
NumPy	Performs numerical operations on loan-duration values.
Pandas	Stores records in tabular form and performs grouping and summaries.
Matplotlib	Creates category-wise and monthly borrowing graphs.
Python IDLE	Used to write and execute the Python program.
GitHub	Hosts the project source files and documentation.

The external Python packages required by the project are listed in requirements.txt: numpy, pandas and matplotlib.

3.1 Hardware Requirements

- A desktop computer or laptop.
- Sufficient memory to run Python and the required libraries.
- Keyboard and display for menu interaction.

3.2 Software Requirements

- Windows, Linux or macOS operating system.
- Python 3 installed.
- NumPy, Pandas and Matplotlib installed.
- Python IDLE or another Python editor.
- A spreadsheet application (optional) for viewing and editing CSV records.
- A web browser and internet connection for accessing the GitHub repository.

4. System Design and Methodology

4.1 Input Data and Schema

The application uses a CSV file named library_records.csv as its primary data source. Each row represents one borrowing transaction and contains a loan identifier, student identifier, book identifier, book title, category, issue date, due date and return date. The return_date field is left blank when a book has not yet been returned.

Field	Description	Example
loan_id	Unique identifier for a loan transaction.	L001
student_id	Identifier of the student who borrowed the book.	S101
book_id	Identifier of the borrowed book.	B101
book_title	Title of the borrowed book.	Python Basics
category	Subject or category of the book.	Technology
issue_date	Date on which the book was issued.	2026-01-05
due_date	Date by which the book should be returned.	2026-01-19
return_date	Actual return date; blank if not returned.	2026-01-15

Dates are represented in YYYY-MM-DD format.

4.2 Data Preparation

At startup, the program checks whether the CSV data file exists. If it is not available, the program creates the required file with demonstration records. The available records are then loaded for display, searching and analysis. The CSV file can also be inspected or edited using spreadsheet software.

4.3 Analysis Workflow

1. Start the program and load or create the CSV file.
2. Read the library borrowing records.
3. Display the main menu and accept the user's selection.
4. Execute the selected operation, such as search, duration calculation or summary generation.
5. Display the result or save the generated graph/report.
6. Return to the menu until the user selects Exit.

4.4 Analytical Measures

Loan duration is calculated as the difference between the issue date and the return date. For a book that has not yet been returned, the current date is used to calculate the elapsed duration. An unreturned record is treated as overdue when its due date is earlier than the current date. Pandas grouping is used to summarise records by category and month, while NumPy is used for numerical calculations on duration values.

5. Implementation of Major Modules

Module / Component	Functionality
LoanRecord class	Represents an individual borrowing transaction and organises its record fields.
CSV handling	Creates, loads and saves library records in library_records.csv.
Add record	Accepts details of a new borrowing transaction and saves the record.
Display and search	Displays all records and searches relevant fields such as loan ID, student ID, book ID, title and category.
Loan-duration analysis	Calculates elapsed borrowing duration and basic numerical statistics.
Overdue checking	Identifies unreturned books whose due dates have passed.
Borrowing frequency	Counts the number of times each book title appears in the loan records.
Category summary	Groups loan records by category and calculates transaction counts.
Monthly summary	Groups issue dates by month and calculates monthly loan counts.
Data-quality checks	Checks duplicate loan identifiers, missing fields and date issues.
Graph generation	Creates a category-wise bar chart and a monthly borrowing trend chart using Matplotlib.
JSON export	Writes a structured summary report to library_report.json.

5.1 Menu Structure

Option	Operation
1	Add a new borrowing record
2	Display all loan records
3	Search records
4	Calculate loan durations
5	Check overdue books
6	Analyse borrowing frequency
7	Generate category-wise summary
8	Generate monthly borrowing summary
9	Check duplicate and incomplete records
10	Generate graphs
11	Export report to JSON
12	Exit the application

5.2 Graphical Output

The application generates two graphical outputs: a bar chart showing the number of loans in each book category and a line chart showing the number of loans by month. The graph images are saved in the graphs directory.

6. Results and Discussion

The application provides a command-line menu through which users can access the implemented functions. The project files include the Python source code, CSV dataset, JSON report, requirements file, README documentation and a graphs folder.

6.1 Application Output

The menu allows the user to add, display and search borrowing records, calculate loan durations, check overdue books, view borrowing frequency and generate category-wise and monthly summaries. It also provides data-quality checking, graph generation and JSON report export.

```
Welcome to the Library Dashboard!

================================================
 LIBRARY CIRCULATION AND DEMAND DASHBOARD
================================================
1. Add a New Loan Record
2. Display All Loan Records
3. Search Loan Records
4. Calculate Loan Durations
5. Check Overdue Books
6. Analyse Borrowing Frequency
7. Generate Category-wise Summary
8. Generate Monthly Borrowing Summary
9. Detect Duplicate and Incomplete Records
10. Generate Graphs and Charts
11. Export Report to JSON
12. Exit
================================================
Enter your choice (1-12):
```

Figure 1. Main menu of the Library Circulation and Demand Dashboard.

```
--- ALL LIBRARY RECORDS ---
loan_id student_id book_id              book_title   category issue_date   due_date return_date
   L001       S101    B101           Python Basics Technology 2026-01-05 2026-01-19  2026-01-15
   L002       S102    B102       The Blue Umbrella    Fiction 2026-01-10 2026-01-24  2026-01-22
   L003       S103    B103     Concepts of Physics    Science 2026-02-02 2026-02-16  2026-02-12
   L004       S104    B101           Python Basics Technology 2026-02-08 2026-02-22  2026-02-20
   L005       S105    B104 A Brief History of Time    Science 2026-03-01 2026-03-15  2026-03-12
   L006       S101    B105           Wings of Fire  Biography 2026-03-10 2026-03-24  2026-03-23
   L007       S106    B102       The Blue Umbrella    Fiction 2026-04-03 2026-04-17  2026-04-14
   L008       S107    B106     Digital Electronics Technology 2026-04-10 2026-04-24
```

Figure 2. Display of all library loan records.

```
Enter Loan ID, Student ID, Book ID or Book Title: L004

--- ALL LIBRARY RECORDS ---
loan_id student_id book_id    book_title   category issue_date   due_date return_date
   L004       S104    B101 Python Basics Technology 2026-02-08 2026-02-22  2026-02-20
```

Figure 3. Search results for a selected record.

```
--- LOAN DURATION ANALYSIS ---
L001 : 10 days
L002 : 12 days
L003 : 10 days
L004 : 12 days
L005 : 11 days
L006 : 13 days
L007 : 11 days
L008 : 213 days
Average Duration: 36.5 days
Shortest Duration: 10 days
Longest Duration: 213 days
```

Figure 4. Loan-duration calculation output.

```
--- OVERDUE BOOKS ---
Digital Electronics | Student: S107 | Days Overdue: 199
```

Figure 5. List of overdue books.

```
--- BOOK BORROWING FREQUENCY ---
Python Basics : 2 times
The Blue Umbrella : 2 times
Concepts of Physics : 1 times
A Brief History of Time : 1 times
Wings of Fire : 1 times
Digital Electronics : 1 times

Most Borrowed Book: Python Basics
Total Borrowings: 2
```

Figure 6. Borrowing frequency analysis.

```
--- CATEGORY-WISE SUMMARY ---
category
Biography     1
Fiction       2
Science       2
Technology    3

Highest Demand Category: Technology
```

Figure 7. Category-wise borrowing summary.

```
--- MONTHLY BORROWING SUMMARY ---
month
2026-01    2
2026-02    2
2026-03    2
2026-04    2
Freq: M
```

Figure 8. Monthly borrowing summary.

```
--- DATA QUALITY REPORT ---
Duplicate Loan IDs: None
Incomplete Records: None
Invalid Dates: None
```

Figure 9. Data-quality check output.

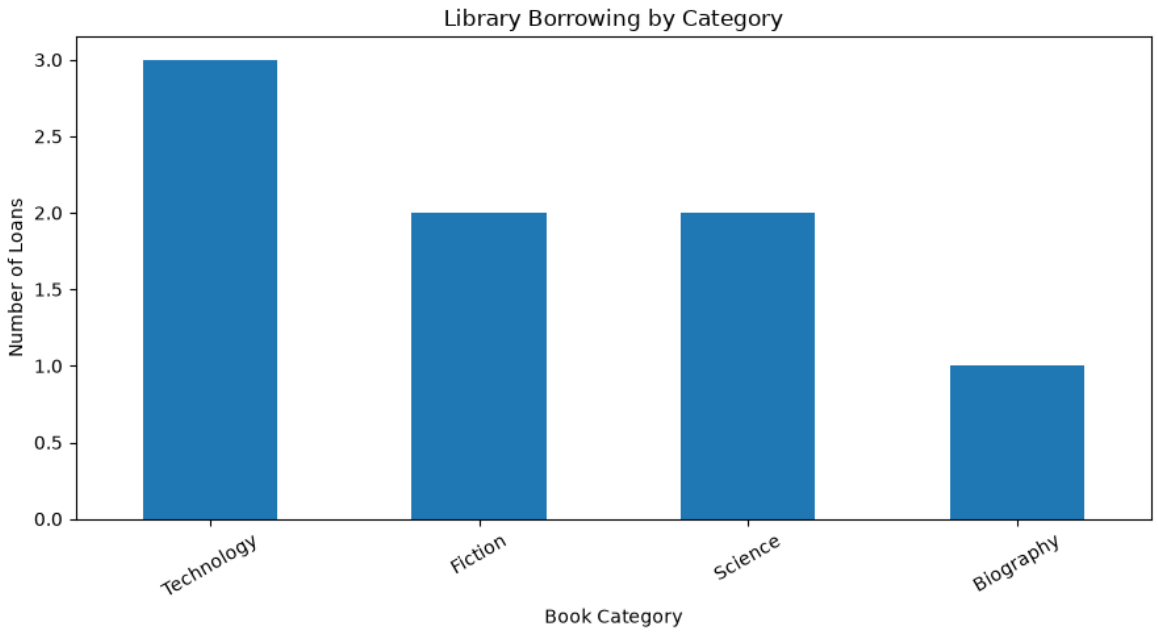


Figure 10. Category-wise borrowing graph (category_analysis.png).

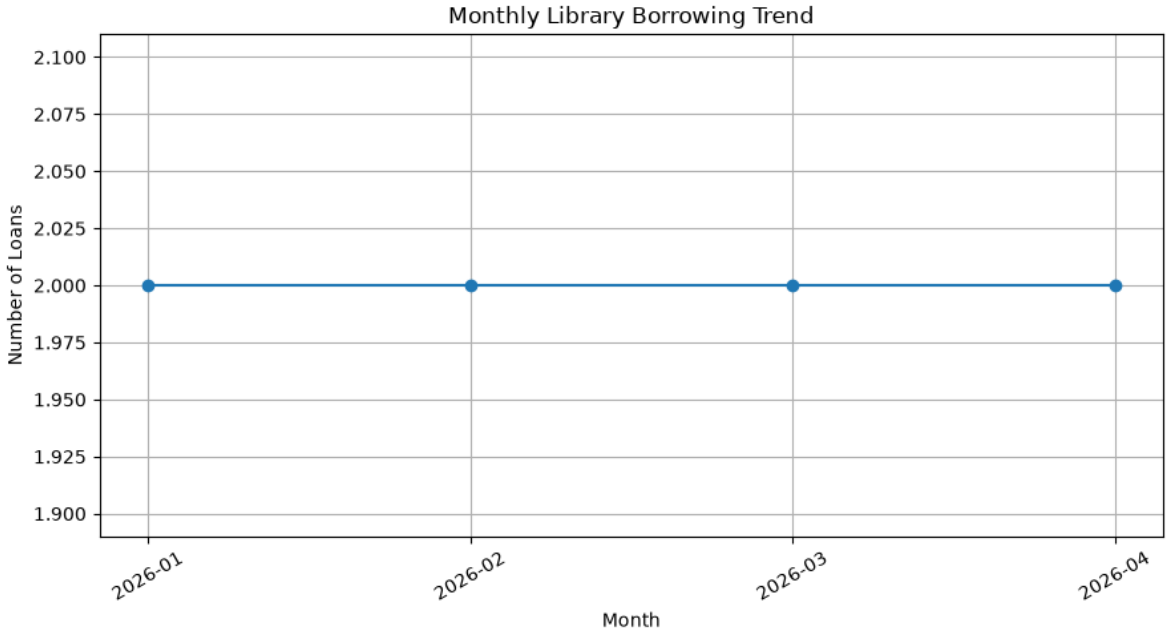


Figure 11. Monthly borrowing trends graph (monthly_trends.png).

```
{
    "generated_on": "2026-11-09",
    "total_loan_records": 8,
    "category_summary": {
        "Biography": 1,
        "Fiction": 1,
        "Science": 2,
        "Technology": 3
    },
    "monthly_summary": {
        "2026-01": 2,
        "2026-02": 2,
        "2026-03": 2,
        "2026-04": 2
    },
    "loan_records": [
        {
            "loan_id": "L001",
            "student_id": "S101",
            "book_id": "B101",
            "book_title": "Python Basics",
            "category": "Technology",
            "issue_date": "2026-01-05",
            "due_date": "2026-01-19",
            "return_date": "2026-01-15"
        },
        {
            "loan_id": "L002",
            "student_id": "S102",
            "book_id": "B102",
            "book_title": "The Blue Umbrella",
            "category": "Fiction",
            "issue_date": "2026-01-10",
            "due_date": "2026-01-24",
            "return_date": "2026-01-22"
        },
        {
            "loan_id": "L003",
            "student_id": "S103",
            "book_id": "B103",
            "book_title": "Concepts of Physics",
            "category": "Science",
            "issue_date": "2026-02-02",
            "due_date": "2026-02-16",
            "return_date": "2026-02-12"
        },
        {
            "loan_id": "L004",
            "student_id": "S104",
            "book_id": "B101",
            "book_title": "Python Basics",
            "category": "Technology",
            "issue_date": "2026-02-08",
            "due_date": "2026-02-22",
            "return_date": "2026-02-20"
        },
        {
            "loan_id": "L005",
            "student_id": "S105",
            "book_id": "B104",
            "book_title": "A Brief History of Time",
            "category": "Science",
            "issue_date": "2026-03-01",
            "due_date": "2026-03-15",
            "return_date": "2026-03-12"
        },
        {
            "loan_id": "L006",
            "student_id": "S101",
            "book_id": "B105",
            "book_title": "Wings of Fire",
            "category": "Biography",
            "issue_date": "2026-03-10",
            "due_date": "2026-03-24",
            "return_date": "2026-03-23"
        },
        {
            "loan_id": "L007",
            "student_id": "S106",
            "book_id": "B102",
            "book_title": "The Blue Umbrella",
            "category": "Fiction",
            "issue_date": "2026-04-03",
            "due_date": "2026-04-17",
            "return_date": "2026-04-14"
        },
        {
            "loan_id": "L008",
            "student_id": "S107",
            "book_id": "B106",
            "book_title": "Digital Electronics",
            "category": "Technology",
            "issue_date": "2026-04-10",
            "due_date": "2026-04-24",
            "return_date": ""
        }
    ]
}
```


Figure 12. Generated JSON report (library_report.json).

7. Testing and Validation

The application was executed in Python IDLE and the menu options were checked using the project CSV data. The following test cases provide a structured record of the expected behaviour. The team observed result on the final demonstration system before submission.

Test case	Expected behaviour	Observed / status
Launch application	CSV is loaded and the main menu is displayed.	Working (team-tested)
Display records	All available records are displayed in tabular form.	Working (team-tested)
Search record	Matching record(s) are displayed for a valid search.	Working (team-tested)
Calculate duration	Loan duration values/statistics are displayed.	Working (team-tested)
Check overdue books	Unreturned loans past their due dates are listed.	Working (team-tested)
Borrowing frequency	Book-title borrowing counts are displayed.	Working (team-tested)
Category summary	Loan counts are grouped by category.	Working (team-tested)
Monthly summary	Loan counts are grouped by month.	Working (team-tested)
Data-quality check	Duplicate or incomplete records are reported.	Working (team-tested)
Generate graphs	Category and monthly graph images are saved.	Working (team-tested)
Export JSON	A JSON report file is created.	Working (team-tested)
Add new record	A valid record is saved to the CSV file.	Verified

8. Limitations and Future Scope

8.1 Limitations

- The application uses a local CSV file rather than a central database.
- It does not include user login, role-based permissions or audit history.
- The project does not manage the complete book catalogue, stock count or reservations.
- Concurrent editing by multiple users is not supported.
- Analysis quality depends on the completeness and correctness of the input records.
- The current interface is text-based and does not provide a graphical user interface.
- Automatic email, SMS or application notifications are not included.

8.2 Future Scope

- Replace CSV storage with SQLite or another database for more reliable data management.
- Develop a graphical user interface or web-based dashboard.
- Add book inventory management, availability tracking and reservation features.
- Introduce login facilities and separate librarian and student roles.
- Provide automated reminders for books approaching or passing their due dates.
- Add filters for date ranges, categories, students and book titles.
- Expand the analysis with demand forecasting when a sufficiently large and reliable dataset is available.
- Deploy the application on a shared server for access by authorised users.

9. Conclusion

The Library Circulation and Demand Dashboard demonstrates the use of Python programming for organising and analysing library borrowing records. The application combines file handling, object-oriented representation of loan records, date calculations, NumPy operations, Pandas-based summaries, Matplotlib visualisations and JSON report export.

Through its menu-driven design, the system allows users to add, display and search records, calculate borrowing durations, identify overdue loans, examine borrowing frequency and generate category-wise and monthly summaries. It also includes basic data-quality checks and graphical reporting. The project provided practical experience in integrating Python libraries and developing a small data-analysis application. The current version serves as an academic prototype that can be extended with a database, a graphical interface and additional library-management features.

10. References

- Python Software Foundation. Python 3 Documentation. https://docs.python.org/3/
- NumPy Developers. NumPy Documentation. https://numpy.org/doc/
- pandas Development Team. pandas Documentation. https://pandas.pydata.org/docs/
- Matplotlib Development Team. Matplotlib Documentation. https://matplotlib.org/stable/
- GitHub Docs. https://docs.github.com/

11. Appendix: CSV Format and Execution

The project uses library_records.csv to store borrowing transactions. The CSV file contains the following field names:

loan_id, student_id, book_id, book_title, category, issue_date, due_date, return_date

Example format (illustrative):

loan_id	student_id	book_id	book_title	category	issue_date	due_date	return_date
L001	S101	B101	Python Basics	Technology	2026-01-05	2026-01-19	2026-01-15
L002	S102	B102	The Blue Umbrella	Fiction	2026-01-10	2026-01-24	2026-01-22

The example rows above explain the CSV structure; retain the actual demonstration records supplied with the project for execution.

11.1 Execution Instructions

7. Install Python 3 on the computer.
8. Open Command Prompt or a terminal in the project directory.
9. Install the required packages using: pip install -r requirements.txt
10. Ensure that main.py and library_records.csv are in the same project directory.
11. Open main.py in Python IDLE and select Run → Run Module (F5), or execute: python main.py
12. Select an option from the displayed menu. The program creates or updates output files as required.
13. The graphs directory stores the generated graph images, and library_report.json stores the exported report.

11.2 Project Repository

The complete source code and supporting files are maintained in the team's GitHub repository:

https://github.com/srijani-dn/Library-Circulation-and-Demand-Dashboard